CHANDU PRASAD BHAIRAPU



AI Developer

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ABOUT ME

Master's graduate in Artificial Intelligence with 3 years of professional experience in AI engineering and software development. Specialized in Python engineering for AI systems, from automated evaluation and CI/CD-integrated checks to containerized deployment on AWS and Azure. Most recently sole developer of an agentic knowledge-extraction system for the German energy sector, backed by a strong foundation in APIs, data analysis and maintainable, well-tested code across enterprise and start-up environments.

EXPERIENCE

Associate Generative AI Developer - Agentic Indexing Pipeline Engineer for German Energy Sector

Kniit GmbH

November 2025 – present Muldestausee, Germany

- Sole developer owning an agentic GenAI knowledge-extraction system end-to-end: data pipelines for ingestion, chunking, prompt engineering and indexing (agentic, deterministic and monolithic strategies), reporting directly to the CTO and working with SAP domain experts across the German energy sector.
- Designed and implemented a Model Context Protocol (MCP) query service that connects the knowledge graph to AI agents with millisecond-latency retrieval, integrated directly into Claude Code for agentic development workflows.
- Designed and optimized multi-agent LLM workflows with LangGraph (entity-extraction, relationship-analysis, consolidation and reflection agents running in parallel) on complex regulatory and technical documents, validating the output against an SAP-system-derived ground truth and raising precision/recall from 33%/45% to 93%/90%.
- Developed Expert Interface, a graph curation web app (React frontend, Python microservices backend) deployed on Azure, where domain experts review and approve knowledge-graph changes before they are applied.
- Engineered a deterministic (code-based) extraction pipeline for a second, high-volume source stream, replacing LLM calls with testable code, cutting a $200,000 projected LLM budget to near-zero recurring cost while pushing extraction accuracy toward 100%.
- Delivered as an AI-augmented engineer, using Claude Code, custom skills and agentic coding workflows for rapid, high-quality development grounded in clean code principles.
- **Tech stack: Python, Git, Docker, Kubernetes, Azure, FastAPI, REST APIs, LangGraph, LangChain, LiteLLM, MCP, RAG, Postgres, Neo4j, MongoDB, React**

GenAI Developer (AI Consulting Intern) - Development and Deployment of an Agentic AI Workflow

Covestro Deutschland AG

December 2024 – April 2025 Leverkusen, Germany

- Worked with business stakeholders to map their decision-making workflows and translated the requirements into an agentic AI application with LangGraph, deployed to AWS, that routes each request across vector database queries, general LLM inference and summarization.
- Developed a Python backend with FastAPI REST APIs and a React front-end, using Postman for API testing.
- Built and containerized the entire stack with Docker and deployed it to AWS using CDK (infrastructure as code), following clean code practices; version control in GitLab with a GitLab CI/CD pipeline for deployment.
- **Tech stack: Python, GitLab CI/CD, Git, Docker, AWS (CDK, EC2, Lambda), FastAPI, Postman, LangGraph, LangChain, React**

GenAI Developer (Master's Thesis) - Evaluating the Performance of RAG Systems

Covestro Deutschland AG

May 2024 – November 2024 Leverkusen, Germany

- Automated evaluation of RAG generation and retrieval with RAGAS and PytrecEval, comparing model answers to chemist-provided ground truth.
- Designed RAG pipelines, focusing on chunking strategies, vector databases, indexing, and fine-tuning models for optimized retrieval.

- Built AWS cloud infrastructure with Lambda, SQS and DynamoDB for PrismaChat RAG evaluation using CDK.
- Benchmarked multilingual embedding models on over 1M data chunks of Covestro's internal chemical data.
- **Tech stack: Python, AWS (Lambda, SQS, DynamoDB, CDK), Docker, RAGAS, PytrecEval, LangChain, Hugging Face Transformers, PyTorch**

Data Scientist (Intern) – ML-Based Prediction of Prepolymer Synthesis Outcomes

Covestro Deutschland AG

November 2023 – April 2024 | Leverkusen, Germany

- Built and compared Random Forest, XGBoost and LSTM models to predict prepolymer synthesis outcomes (e.g. viscosity, reaction progression) from time-series process data, evaluating tree-based vs. deep learning approaches for this tabular/temporal hybrid problem.
- Engineered lag features and rolling statistical windows from raw sensor time-series (temperature, reaction time, process variables) to feed the tree-based models, while feeding raw sequences directly into the LSTM.
- Benchmarked model performance against traditional empirical curve-fitting methods, improving early identification of batches likely to deviate from target specification.
- Deployed an interactive Streamlit dashboard exposing model predictions to chemists in real time, letting them simulate parameter changes and see predicted outcomes before running physical batches.
- **Tech stack: Python, Scikit-Learn, XGBoost, TensorFlow/Keras, Pandas, NumPy, Streamlit**

Machine Learning Engineer - Electrical Vehicle Load Forecasting using Time Series Analysis

Cognizant Technology Solutions

September 2021 – March 2022 | Hyderabad, India

- Developed ARIMA, SARIMA, XG-Boost regressor, FB Prophet, LSTM models for time series analysis for EVs.
- Developed LSTM, achieved 96% test accuracy and strong performance on unseen data.
- **Tech stack: Python, Scikit-Learn, TensorFlow**

PROJECTS

Project: AI Output Quality Gates | Python | pytest | GitLab CI/CD | Docker

- Python quality engineering framework and CLI that runs golden-set, schema and regression checks on LLM and agent outputs, with a plugin interface for team-owned checks and clear diagnostics per failure.
- Runs as a GitLab CI/CD job in a Docker image on every merge request and fails the pipeline on regressions, with docs and example checks.

TECHNICAL SKILLSET

- **Programming & Delivery:** Python, Git, CI/CD pipelines (GitLab CI/CD), containers (Docker), Postman, FastAPI, clean code, Agile
- **Cloud & Infrastructure:** AWS (CDK for infrastructure as code, Lambda, SQS, DynamoDB, EC2), Azure, Kubernetes
- **Evaluation & GenAI:** LLM and RAG evaluation (RAGAS, PytrecEval), LangGraph, LangChain, MCP, RAG, Claude Code
- **Data:** SQL, data pipelines, data validation, Postgres, MongoDB, Neo4j, Vector Databases (Qdrant, Pinecone), Pandas, Scikit-Learn, PyTorch
- **Languages:** English (Business Fluent), German (B1), Hindi (Fluent), Telugu (Native)

EDUCATION

Master of Science in Artificial Intelligence — March 2022 – April 2025
Technische Hochschule Würzburg-Schweinfurt, Würzburg, Germany

Bachelor of Technology in Mechanical Engineering — August 2017 – July 2021
G Pulla Reddy Engineering College, Kurnool, India (Minor in Computer Science & Engineering)

PUBLICATIONS & CERTIFICATIONS

- **[1]** Navigating the Future: An approach of Autonomous Indoor Vehicles
- Udemy - LLMs Mastery : Transformers & Generative AI